

Agentic Systems: Builders, Architectures & the Research Frontier

A technical survey of who is building agentic software and how — runtimes, protocols, benchmarks, and in-flight research — with three deep dives and a comparison. Assumes familiarity with LLMs, tool-calling, and RL.

Compiled 2026-05-26. GitHub stars fetched live this week. Benchmark scores and private valuations move fast; confidence is flagged inline (H verified / M reputable secondary / L unverified). arXiv IDs verified against arxiv.org unless flagged.

TL;DR for engineers

The stack is consolidating into layers: **(1) frontier models** with native tool-use/computer-use + RLVR post-training; **(2) agent runtimes/frameworks** (LangGraph, OpenAI Agents SDK, MS Agent Framework, CrewAI, Pydantic AI...); **(3) interop protocols** — **MCP** (agent↔tools) + **A2A** (agent↔agent), with agentic-payment standards still fragmented; **(4) products** (coding agents, computer-use, vertical enterprise).

Key shifts since 2025: SWE-bench Verified is saturating (→ **SWE-bench Pro**, contamination-resistant); evals moving from one-shot to **pass^k reliability** (τ-bench); **computer-use (OSWorld)** and **long-horizon** autonomy remain the hard walls; **RLVR** is the dominant training recipe (with reward-hacking as the live failure mode); memory is moving from RAG-over-history to **agent-managed self-editing memory** (MemGPT/Letta, Mem0, A-MEM).

Contents

1. Taxonomy & reference architecture
 2. Coding agents & IDEs
 3. Frameworks, runtimes & infra
 4. Computer-use & vertical agents
 5. Three deep dives (Cursor / Manus / Sierra)
6. Technical comparison
 7. Interop & payment protocols
 8. Benchmarks & evals
 9. Research frontier (incl. in-progress)
 10. Open problems & what to watch

1 · Taxonomy & reference architecture

An "agentic" system = an LLM in a **perceive→plan→act→observe** loop with tools, memory, and some autonomy over control flow. Minimum viable agent: model + tool schema + loop + stop condition. Production systems add the layers below.

```

# Reference layering (bottom → top)
[ Model ]      frontier LLM w/ tool-use, reasoning/extended-thinking, computer-use
[ Harness ]    agent loop · context mgmt/compaction · planner · reflection · verifier
[ Memory ]     short-term (context) + long-term (vector/graph) + self-editing (MemGPT)
[ Tools ]      function-calling · code exec (E2B sandbox) · browser · file I/O · MCP servers
[ Interop ]    MCP (agent↔tools) · A2A (agent↔agent) · AGNTCY (discovery/infra)
[ Orchestr.]   single-agent | orchestrator-worker | role-based multi-agent | swarm
[ Eval/Obs ]   traces · benchmarks · pass^k reliability · LangSmith/Logfire/OTel
  
```

Two orthogonal axes that matter most in practice: (a) *autonomy granularity* — atomic tools the model composes freely vs. hard-coded workflows (the "agent-native" vs. pipeline debate); (b) *reliability surface* — guarantees must live in tool/harness code, not prompts (prompts are suggestions). Token cost scales ~linearly with autonomy: Anthropic reports multi-agent orchestration at ~15× single-agent tokens. M

2 · Coding agents & IDEs

The most commercially mature category. Ranked by scale (valuation/ARR for closed; stars for OSS). Stars fetched 2026-05-26.

Tool	Architecture / approach	GitHub ★	Scale	Pricing	OSS	Link
Cursor	VS Code fork; agent + background agents; custom retrieval/"context engine"	closed	~\$29.3B¹ · ~\$2B ARR	\$0/\$20/\$60/\$200	no	site
Claude Code	Terminal harness; agentic file/test loop; Agent SDK; MCP-native	~127k	Anthropic ~\$30B run-rate¹	Claude plans / API	proprietary	repo
Devin (Cognition)	Cloud autonomous SWE; planner + shell + browser + editor; ACU billing	closed	~\$10.2B¹ · ~\$73M ARR	\$20 PAYG / \$500 / ent	no	site

Gemini CLI	Open terminal agent; ReAct loop; MCP; large free tier	~105k	Google-backed	Free / API	Apache-2.0	repo
Codex (OpenAI)	Terminal + cloud agent; Responses API; sandboxed exec	~86k	via ChatGPT	ChatGPT / API	Apache-2.0	repo
Zed	Rust editor; native agent panel; high-perf collab	~84k	VC-backed	Free + AI usage	GPL/AGPL	repo
OpenHands	Agent + sandboxed runtime; CodeAct-style; SDK/CLI/GUI/cloud	~75k	~\$24M raised	OSS + cloud	MIT	repo
Cline	VS Code ext; Plan&Act modes; BYO keys; MCP	~62k	high traction	BYO model	Apache-2.0	repo
Replit Agent	Cloud IDE; full app build/deploy; checkpoint billing	closed	~\$9B · ~\$150M ARR	\$0/\$20/\$100	no	site
Aider	Terminal pair-programmer; git-aware diff/commit loop; repo map	~45k	indie	Free (your API)	Apache-2.0	repo
Windsurf (Cognition)	Agentic IDE "Cascade"; acq. by Cognition Jul'25	closed	part of Cognition	\$0/\$20/\$200	no	site
Factory	"Droids"; agent-native full SDLC; terminal + cloud	closed	~\$1.5B	\$0/\$20/ent	no	site

Also: Continue (~33k, OSS IDE assistant + CI review), Goose (~46k, Block, MCP-first local agent), Augment (closed, large-context enterprise), Amp (Sourcegraph), Tabnine (privacy/air-gapped). **Dead/archived:** Roo Code (archived May'26), Aide/CodeStory (archived Feb'25), OpenAI Swarm (deprecated → Agents SDK).

3 · Frameworks, runtimes & infra

The build-your-own layer. Pattern: OSS library is free; revenue from hosted observability/eval (LangSmith, Logfire), managed runtime (LangGraph Platform, AgentOS, Vertex Agent Engine), or no-code cloud (Dify, n8n). Stars 2026-05-26.

Project	What / primitives	★	License	Backing / size	Status note
LangChain/Graph	General framework; LangGraph = stateful graph orchestration	138k/33k	MIT	\$1.25B val · ~90M dl/mo	de-facto standard
Dify	Visual agentic-workflow platform	143k	Apache*	~\$180M val	strong self-host
n8n	Workflow automation + native AI nodes	190k	fair-code	\$2.5B val	not OSI-OSS
CrewAI	Role-based multi-agent "crews"; tasks/process	52k	MIT	~\$18M raised	popular for multi-agent
MS Agent Framework	Unified successor merging AutoGen + Semantic Kernel	11k	MIT	Microsoft	1.0 GA ~Apr'26
AutoGen	Multi-agent conversation; Magentic-One	58k	MIT	Microsoft	maintenance only
Semantic Kernel	Plugins/planners (.NET-first)	28k	MIT	Microsoft	maintenance only
OpenAI Agents SDK	Agents/Handoffs/Guardrails/Sessions on Responses API	27k	MIT	OpenAI	prod successor to Swarm
LlamaIndex	RAG/knowledge agents; LlamaParse	50k	MIT	~\$27.5M raised	RAG-centric
Pydantic AI	Type-safe agents; structured outputs; Logfire	17k	MIT	Pydantic/Sequoia	strong typing
Google ADK	Code-first multi-lang agent toolkit; Vertex Agent Engine	20k	Apache-2.0	Google Cloud	A2A-aligned
Smolagents	CodeAgent (agents that "think in code")	28k	Apache-2.0	HuggingFace	minimalist
Vercel AI SDK	TS toolkit; streaming/tool-use; ~20M dl/mo	25k	Apache-2.0	Vercel \$9.3B	TS standard
Mastra	TS agent framework (ex-Gatsby team)	24k	Apache-2.0	\$13M seed	rising TS option
Letta (MemGPT)	Stateful agents w/ self-editing tiered memory	23k	Apache-2.0	\$10M seed	memory leader
Agno (Phidata)	High-perf agents + AgentOS runtime	40k	MPL-2.0*	—	perf-focused
Composio	1000+ tool/auth layer; tool search	29k	OSS core*	~\$80M raised	integration layer
E2B	Secure code-exec sandboxes for agents	12k	Apache-2.0	\$32M raised	runtime infra

Also: Haystack (25k, deepset, prod RAG), CAMEL/OWL (17k/20k, multi-agent research + GAIA), SuperAGI (17k). * license has commercial conditions — verify before redistribution.

4 · Computer-use & vertical agents

Computer/browser use

Product	Approach	OSS / scale	Pricing	Link
OpenAI ChatGPT Agent	Virtual-computer agent (absorbed Operator/CUA); vision + RL on GUIs	closed	ChatGPT Plus/Pro; API CUA \$3/\$12 per 1M	site
Anthropic computer use	Screenshot-driven GUI control tool in API; strongest coding models	closed	API per-token; Pro/Max	site
Google Project Mariner	Gemini Chrome agent; ~10 parallel tasks; computer-use in Gemini API	closed	AI Ultra \$249.99/mo	site
Browser Use	OSS framework to drive browsers (DOM + vision); used at Anthropic/Amazon	MIT ~96k★	OSS + cloud API	repo
Skyvern	OSS vision+LLM RPA; selector-free web automation	AGPL ~22k★	OSS + cloud	repo

Vertical enterprise (where the revenue concentrates)

Company	Domain	What the agent does	Scale	Pricing	Link
Sierra	CX	End-to-end customer-service resolution agents	~\$15B · ~\$150M ARR	outcome-based	site
Harvey	Legal	Contract/research/diligence agents	~\$11B · ~\$190M ARR	seat/enterprise	site
Glean	Enterprise search	Search + cross-app action agents	~\$7.2B · ~\$200M ARR	seat + usage	site
Decagon	CX	Concierge support agents	~\$4.5B · ~\$35M ARR	outcome/usage	site
Hebbia	Finance/legal	Multi-agent doc analysis (Matrix)	~\$700M (funding stale)	enterprise	site
Manus	General	Autonomous general agent on a virtual computer	reported Meta acq. \$2B+ ¹	credits	site

5 · Three deep dives

Cursor (Anysphere) — retrieval-first agentic IDE

closed-source · VS Code fork · ~\$2B ARR · ~\$29.3B¹

Technical thesis: the bottleneck for code agents is *context selection*, not raw model quality. Cursor's moat is a proprietary **codebase indexing/retrieval ("context engine")** + tight editor integration + **background agents** (async, parallel, cloud-executed tasks). Model-agnostic: routes across frontier models (Anthropic/OpenAI/its own fine-tunes) and applies edits via fast "apply" models.

- **Loop:** retrieve relevant context → propose multi-file diff → user/agent applies → run/iterate. Agent mode plans across files; background agents run headless.
- **Why it won:** latency + UX on a daily workflow; "Tab" completion + agent in one surface. Fastest-ever B2B \$0→\$2B ARR.
- **Risk:** sits atop frontier-lab models that now ship their own agents (Claude Code, Codex) — margin/differentiation pressure.

Manus (Butterfly Effect / Monica) — general agent on a virtual computer

closed-source · China · ~\$100M ARR in ~8mo · reported Meta acq. \$2B+¹ **L**

Technical thesis: wrap frontier models (reportedly Claude + Qwen-class) in a **persistent cloud VM** with shell, browser, file system, and a planning/"todo" loop, then let the agent run long-horizon tasks asynchronously to a finished deliverable. Emphasizes **transparent execution** ("Manus's Computer" view) and a context-engineering approach to long trajectories (KV-cache-friendly prompts, file-system-as-memory, recitation of goals to combat drift).

- **Self-reported GAIA** (Manus 1.5): ~86% L1 / ~70% L2 / ~58% L3 **L** — treat as marketing until third-party-verified.
- **Significance:** productized the "do the whole job" long-horizon pattern; proof a breakout agent can ship from anywhere and scale in months.
- **Risk:** generalist agents face reliability/cost vs. focused agents; the Meta acquisition price is press-reported, not confirmed.

Sierra (Bret Taylor) — outcome-priced enterprise CX agents

closed-source · ~\$15B · ~\$150M ARR · outcome-based pricing

Technical thesis: reliability + governance are the product, not the model. Sierra wraps models in a **guardrailed, policy-constrained agent** with deterministic tool integrations (CRM, order systems), supervisory checks, and per-customer evals. Co-developed **τ-bench** / **τ²-bench** (Sierra Research) to measure exactly the thing enterprises care about: **multi-turn reliability and policy adherence under pass^k**.

- **Pricing innovation:** charges largely **per resolved outcome** — only viable because the agent does verifiable end-to-end work; aligns vendor incentive with task success.
- **Significance:** sets the template for vertical agents (reliability eng + outcome pricing + domain integrations) and contributes real eval methodology back to the field.
- **Risk:** outcome attribution/guarantees are hard; long enterprise sales cycles; incumbents (Salesforce Agentforce) competing.

6 · Technical comparison

Axis	Cursor	Manus	Sierra
Agent topology	Single-agent + async background agents	Single long-horizon agent on a VM (+ sub-tasks)	Guardrailed single-agent + supervisory checks
Core moat	Codebase retrieval + editor UX + apply models	Long-horizon context engineering + VM tooling	Reliability/governance + domain integrations
Model strategy	Multi-model router + own fine-tunes	Wraps frontier models (Claude/Qwen-class)	Model-agnostic; policy layer on top
Reliability approach	Human-in-loop diff review; run tests	Self-verification, recitation, file memory	Guardrails + evals (τ-bench) + pass^k
Pricing	Flat subscription tiers	Subscription + usage credits	Per-resolution (outcome-based)
Eval exposure	SWE-bench-class (coding)	GAIA (self-reported)	τ-bench (authored)
Failure mode	Context miss → wrong edit	Long-horizon drift / cost blowup	Outcome misattribution / policy gaps

Pattern: differentiation has moved **above the model** — into retrieval (Cursor), long-horizon context engineering (Manus), and reliability/governance + pricing (Sierra). All three treat the frontier model as a swappable component.

7 · Interop & payment protocols

MCP — Model Context Protocol (Anthropic, community-governed)

JSON-RPC 2.0; roles **Host / Client / Server**; servers expose **Resources, Prompts, Tools**; clients expose **Sampling, Roots, Elicitation**. Stateful w/ capability negotiation (modeled on LSP). Ratified spec 2025-11-25 **H**. RC in progress: stateless core, **Extensions, Tasks** (async long-running), **MCP Apps** (UI), OAuth 2.1 auth hardening **M**. Public server registry growing; verified/security-rated registry on 2026 roadmap. De-facto "tool port" for agents (OpenAI/Google interop). [spec](#)

A2A — Agent2Agent (Google → Linux Foundation, Jun 2025)

Agent↔agent (peer), complementary to MCP. Primitives: **Agent Cards** (capability discovery), **Tasks**, transport over HTTP + SSE + JSON-RPC. Opaque-agent interop without exposing internal state. 100+ supporters; vendor-neutral under LF. **H** The emerging two-layer stack = **MCP (tools) + A2A (agents)**. [a2a-protocol.org](#)

AGNTCY / "Internet of Agents" (Cisco + LangChain + Galileo → LF)

Broader infra stack: discovery, composition, deployment, eval; OASF agent manifest/directory. 65+ companies; LF governance. Overlaps/competes with A2A at the infra layer. [agntcy.org](#)

Agentic payments (fragmented — no winner)

Open problems: delegated authorization, scoped consent, agent identity/attestation, dispute/liability. Contenders: **AP2** (Google, signed mandates/verifiable credentials, 60+ partners incl. Mastercard/PayPal/Coinbase), **x402** (Coinbase, HTTP-402 stablecoin micropayments for M2M), **Visa TAP** (agent identity in HTTP headers), **Mastercard Agent Pay** (tokenized agentic credentials), **ACP** (OpenAI/Stripe in-ChatGPT checkout). All **M/L** on adoption specifics.

8 · Benchmarks & evals

Scores move weekly and secondary sources conflict. Treat specific top-line % as snapshots; several "leaderboard" pages with future-dated model names and round scores appear auto-generated and unreliable. Confidence flagged per item.

Benchmark	Measures	Notes / state	Ref
-----------	----------	---------------	-----

SWE-bench (Verified)	Resolve real GitHub issues; patch must pass repo tests; 500 human-validated	Headline coding bench; ~70s–80s% SOTA, saturation M	2310.06770
SWE-bench Pro	Harder, contamination-resistant, longer-horizon SWE	~80%-Verified models drop to ~mid-40s% M	2509.16941
SWE-Lancer	1,400 real Upwork tasks, \$1M payout; incl. managerial	Frontier earns a fraction of \$1M M	2502.12115
GAIA	466 real-world reasoning+multimodal+web+tools, 3 levels	Release gap humans ~92% vs GPT-4+plugins ~15% H	2311.12983
τ-bench / τ²/τ³	Tool-Agent-User CX dialogues; policy adherence; pass^k	Sierra; surfaces multi-turn reliability gap H	2406.12045
TheAgentCompany	Simulated software co.; multi-role professional tasks	Frontier completes a minority end-to-end H	2412.14161
OSWorld	369 real OS tasks (Ubuntu/Win/macOS); execution-based	Canonical computer-use wall; release humans ~72% vs ~12% H	2404.07972
WebArena / VWA	Self-hosted web apps; multi-step browser tasks (+visual)	SOTA rising but <human; top numbers L	2307.13854
BrowseComp	1,266 hard-to-find-fact deep-browsing tasks	Targets deep-research agents H	OpenAI '25
Terminal-Bench 2.0	End-to-end terminal/computer tasks (build/train/serve)	Hard long-horizon, verifiable M	tbench.ai
Cybench	40 CTF tasks; agentic offensive-cyber; unguided success	Used in RSP/dangerous-capability evals M	2408.08926

9 · Research frontier (incl. in-progress)

Memory

MemGPT/Letta (2310.08560) — "LLM as OS": virtual context mgmt, tiered memory, self-editing via function calls. **Mem0** (2504.19413) — extraction→consolidation→retrieval; benchmarked on LoCoMo. **A-MEM** (2502.12110) — Zettelkasten-style interconnected evolving notes; gains on multi-hop. Direction: RAG-over-history → **agent-managed self-evolving memory**; eval still immature. H

Self-improvement / open-ended

Reflexion (2303.11366) — verbal RL: reflect on failures into episodic memory (no weight updates). **Voyager** (2305.16291) — open-ended Minecraft agent w/ automatic curriculum + **executable skill library** + self-verification; foundational for lifelong-learning/skill-library agents. H

RL for agents (RLVR, verifiers, test-time compute)

RLVR (RL w/ Verifiable Rewards) is the dominant post-training recipe — reward from programmatic checks (tests/math/format). Canonical: DeepSeek-R1 GRPO rule-based rewards; OpenAI o-series "scale RL + tool use." Live failure mode: **reward hacking / verifier gaming**. Unifying reasoners+verifiers for test-time scaling (2505.04842); contrarian result that RLVR sharpens sampling more than it adds capability (2506.14245). H on these IDs; 2026-dated IDs from search L — verify before citing.

Multi-agent orchestration & long-horizon

Anthropic multi-agent research system: orchestrator-worker; lead decomposes → parallel subagents (explicit objective/format/tools) → synthesize. +90.2% vs single-agent on internal evals at ~15× tokens; token usage explained ~80% of variance. Heuristics: 1 agent fact-find, 2–4 compare, 10+ complex. M Long-horizon autonomy (error compounding, credit assignment, trajectory repair) remains the central open problem.

Computer-use models & world models

Distinct model class: vision + GUI grounding + RL on UI (OpenAI CUA, Anthropic computer-use, Gemini computer-use). OSWorld is the proving ground; grounding is the bottleneck. **World models for agents** (Genie-line / learned simulators, DeepMind) — sample-efficient training/planning, research-stage.

Frontier products & autonomous science

OpenAI (ChatGPT agent absorbing Operator/CUA; o-series + GPT-5.x for agentic coding/computer-use), Anthropic (Claude Code + Agent SDK + MCP; computer-use), Google DeepMind (Project Mariner ~10 parallel tasks; Project Astra real-time multimodal assistant). **Sakana "AI Scientist" v2** (2504.08066) — agentic tree-search research pipeline; produced the first AI-generated manuscript to pass workshop peer review at ICLR 2025 (avg 6.33) — heavily caveated: hallucinations/overstated novelty (critical eval 2502.14297). M

OSS research-grade repos

AutoGPT (~185k), OpenHands (~75k), OpenManus (~56k), SWE-agent (~19k; introduced the Agent-Computer Interface; mini-swe-agent variant), Voyager (ref impl), Letta (~23k).

10 · Open problems & what to watch

- **Long-horizon reliability** — error compounds over long trajectories; credit assignment and mid-run trajectory repair are unsolved. The real moat is reliability eng, not model access.
- **Computer-use grounding** — OSWorld shows GUI grounding + operational knowledge is far from solved; biggest near-term capability unlock.
- **Eval integrity** — Verified saturation/contamination → Pro + pass^k + contamination-resistant suites. Don't trust single-number leaderboards; measure pass^k on *your* task.
- **RLVR reward hacking** — verifier gaming is the dominant failure of the dominant training recipe; verifier design is becoming its own discipline.
- **Interop & identity** — MCP+A2A converging fast; agent identity, scoped consent, and payments are the unsettled frontier (AP2/x402/TAP).
- **Cost of autonomy** — multi-agent ~15× tokens; orchestration topology is a cost/quality dial, not a free lunch.
- **Memory** — self-editing/evolving memory is promising but under-evaluated; LoCoMo-class benchmarks are early.

Sources, dates & confidence

Compiled 2026-05-26. GitHub stars fetched live this week (popularity, not usage). Private valuations/ARR are point-in-time from press (TechCrunch/CNBC/Bloomberg/VentureBeat) and company posts; ¹ = **reported/in-talks, not confirmed** (Cursor ~\$29.3B & rumored ~\$50B; Cognition ~\$10.2B & rumored ~\$25B; Manus Meta acquisition \$2B+). Benchmark percentages are snapshots and move weekly; specific future-dated leaderboard scores were treated as unreliable and not asserted. arXiv IDs verified against arxiv.org except where flagged **L**.

Key refs: MCP [spec 2025-11-25](#) · A2A [a2a-protocol.org](#) · AGNTCY [agntcy.org](#) · Anthropic multi-agent [engineering blog](#) · SWE-bench [swebench.com](#) · OSWorld [os-world.github.io](#) · τ^2 -bench [github](#) · MemGPT 2310.08560 · Mem0 2504.19413 · A-MEM 2502.12110 · Reflexion 2303.11366 · Voyager 2305.16291 · Sakana AI Scientist v2 2504.08066. Companion **Overview Edition** covers the same landscape for non-engineers.